

Assignment 3.

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- Q] To understand Devops: Principles, Practices and Devops engineer role and responsibilities

Devops is a combination of Development and Operations. It is a culture, set of Practices and tools that improves collaboration between software developers and IT operations team. The main goal of Devops is to deliver Software faster, with better quality, and with continuous improvement.

Principles of Devops.

1. Collaboration and Communication
Development, testing and operations teams work together throughout the Software lifecycle
2. Automation
Repetitive tasks like testing, building deployment and monitoring are Automated. to save time and reduce errors.
3. Continuous Integration
Developers frequently merge code into a shared repository where builds and tests run automatically.

4. Continuous Delivery / Continuous Deployment
Software updates are released quickly and reliably through automated pipelines
 5. Monitoring and feedback.
Applications and infrastructure are continuously monitored to detect issues and improve performance.
 6. Customer centric approach.
focus is on delivering value to users faster with frequent updates and improvement.
 7. Infrastructure as code.
Servers, networks and environments are managed using code instead of manual configuration
- ★ Devops Practices
- Continuous Integration
 - Continuous Delivery.
 - Continuous Deployment.
 - Version Control.
 - Automated Testing
 - Configuration Management.
 - Containerization
 - Orchestration
 - Monitoring and logging

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Role of DevOps Engineers

A DevOps Engineer is responsible for bridging the gap between development and operations teams. They automate workflows, manage infrastructure and ensure reliable software delivery. They work with Developers, testers security teams and cloud teams to maintain smooth release cycles.

Responsibilities of a DevOps Engineer
Build and Replace management: Create automated build and deployment pipelines

CI/CD pipelines setup: Use tools like Jenkins Github Actions or Github CI/CD

Infrastructure Management: Provision and Maintain servers, cloud resources and environment

Automation: Automate testing, deployments backups and repetitive tasks.

Monitoring and Incident Responses.

Security Integration (DevOps) Include security checks in Pipelines and Infrastructure.

Collaboration: Coordinate between developers, QA, and operations teams.

Performance Optimization: improve scalability, reliability etc.

Documentation: Maintain runbooks, deployment guides and architecture documents.